Amirhossein Khalilian-Gourtani

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Personal Profile

Amirhossein is a Ph.D. student in electrical engineering at New York University Tandon school of engineering. Received his M.Sc. degree from New York University in electrical engineering. His main research interests include Signal Processing, Machine Learning, Convex and Non-smooth Optimization, Medical Signal Processing, and Numerical Analysis.

Education

Sep.2018–Present **Ph.D. in Electrical Engineering**
New York University, NY, USA.
- GPA: **4.0/4.0**

Sep.2016–Aug.2018 **M.Sc. in Electrical Engineering**
New York University, NY, USA.
- GPA: **4.0/4.0**

Sep.2012–Aug.2016 **B.Sc. in Electrical Engineering (Communications)**
Isfahan University of Technology, Isfahan, Iran.
- GPA:**19.21/20**

Honors and Awards

2018 **Awarded** the **Myron M. Rosenthal Award** for **Best M.Sc. Academic Achievment in Electrical and Computer Engineering** from **New York University**.

2017 **Awarded** the **Samuel Morse M.Sc. Fellowship** from **New York University**.

2017 **Awarded** the **Certificate of Merit** for **Academic Achievement** from **New York University Tandon School of Engineering**.

2016 **Awared** the **Graduate School of Eng. Scholarship** from **New York University.**.

2015 **Award** and **Scholarship** from **National Elites Foundation,** governed by Presidency of Islamic Republic of Iran.

Ranked 2nd, among undergraduate students of ECE Department, Isfahan University of Technology, Class 2016.

Ranked 1st, among students of Communication Engineering, ECE Department, Isfahan University of Technology, Class 2016.

2013-2015 **Awarded as an exceptionally talented student for 3 consecutive years ,** by Isfahan University of Technology.

Research Interests

• Convex and Non-smooth Optimization • Signal Processing • Machine Learning
• Medical Signal Processing • Numerical Analysis

Publications

- R. Wang, A. Khalilian-Gourtani, et. al. , "Stimulus Speech Decoding from Human Cortex with Generative Adversarial Network Transfer Learning," 2020 IEEE International Symposium on Biomedical Imaging (ISBI 20).
- A. Khalilian-Gourtani, M. Tepper, V. Minden D. Chklovskii, "STRIP THE STRIPES: Artifact Detection and Removal for Scanning Electron Microscopy Imaging," 2019 44th International Conference on Acoustics, Speech, and Signal Processing (ICASSP).
- A. Khalilian-Gourtani, Y. Wang and J. Mamou, "Scanning Acoustic Microscopy Image Super-Resolution using Bilateral Weighted Total Variation Regularization," 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC).
- Khoroushi M, Sheikhi M, Khalilian-Gourtani A, Soleimani B. Effect of root canal rinsing protocol on dentin bond strength of two resin cements using three different method of test. J Clin Exp Dent. 2016..

Patents

- A.Khalilian-Gourtani, N. Gadgil, G.-M. Su, “User-guided image segmentation methods and product,” US Patent Application 62/944,847, Assignee: Dolby Laboratories Inc., November 2019.

Professional Experiences

Summer of 2019	Video Processing Research Intern, Dolby Labratories Inc..
Summer of 2018	Summer Research Associate, Center of Computational Biology, Simons Foundation-Flatiron Instutue.
Spring of 2018	Teacher Assistant, Image and Video Processing, NYU Tandon School of Engineering, Lecturer: Prof. Wang.
Fall of 2017	Teacher Assistant, Medical Imaging, NYU Tandon School of Engineering, Lecturer: Dr. Mamou.
2017	Volunteer Research Assistant, Langone Health NYU.
Spring of 2016	Teacher Assistant, Data Communications and Computer Networks, Isfahan University of Thechnology, Lecturer: Dr. Naghsh.
Spring of 2016	Teacher Assistant, Circuit Theory I, Isfahan University if Technology, Lecturer: Prof. Saeedi.
Spring of 2015	Teacher Assistant, Signals & Systems Analysis, Isfahan University of Technology, Lecturer: Dr. Narimani.
Summer of 2014	Cooperation with SarvNet Telecommunication Inc on Design and Implementation of an AES FPGA core.

Professional skills

Programming Languages	MATLAB & SIMULINK, Python, Verilog, AVR Codevision, C, Julia
CAD Tools	Orcad Pspice, CST Studio, Cisco Packet Tracer, Proteus
Miscellaneous	Microsoft Office, LaTeX, Fluent English, Teamwork skills, Learn quickly